

AUTO DATA STORYTELLER REPORT

Executive Summary

Here is the Executive Summary: Our recent analytical effort has successfully developed a highly accurate predictive model, capable of correctly classifying over 83% of records. This robust performance indicates a strong understanding of underlying patterns and provides a reliable foundation for data-driven decision-making. The analysis reveals that gender, particularly male, is the most significant factor influencing predicted outcomes. The amount paid (fare) and age also emerged as crucial determinants. These findings suggest that understanding demographics and pricing strategies are paramount. We recommend close, ongoing monitoring of these key variables, as shifts in their patterns are likely to have the most substantial impact on future business results and strategic planning.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

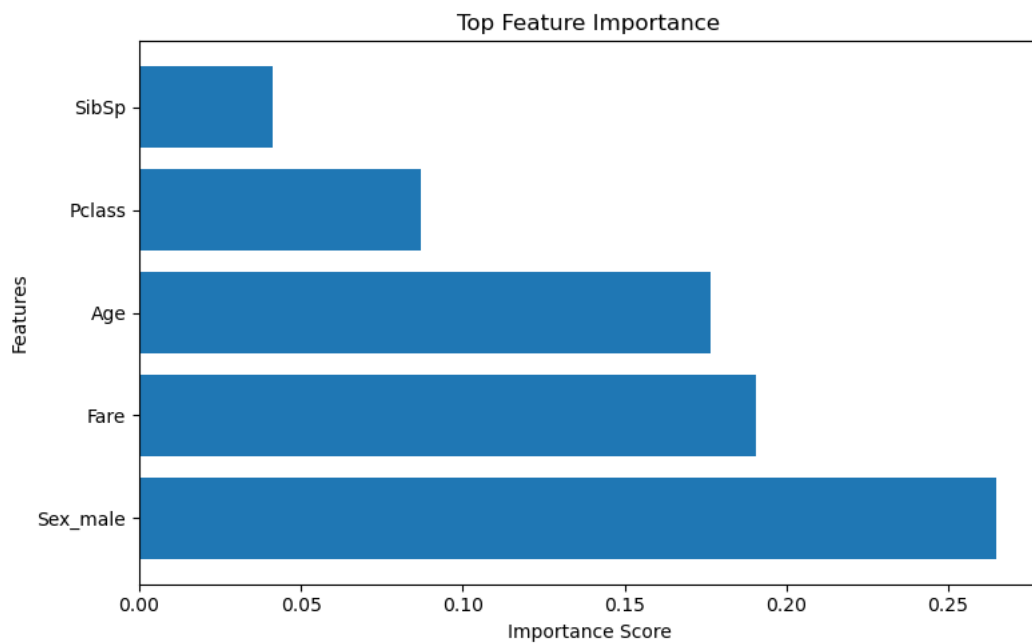
Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.

- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

